Foundations of Programming

Assignment No.: 17

PROBLEM STATEMENT: To accept the number of terms and find the sum of the sine series.

OBJECTIVE:

- To understand the concept of mathematical series in programming.
- To implement sine series using C programming.
- To practice loops and arithmetic operations.
- To learn the use of factorial and power calculations in C.

THEORY:
In mathematics, the sine function can be expressed as an infinite series called the Taylor series expansion. The sine series is represented as:
sin(x) = x − x^3/3! + x^5/5! − x^7/7! + ...

In this program, the user enters the value of x and the number of terms. The program calculates the sum of the sine series up to the specified number of terms using loops. Factorial and power operations are used to compute each term of the series.

PLATFORM: Linux

CODE:

```
#include <stdio.h>

#include <math.h>

int main()

{

int terms,i,sign=1;

float x,sum=0,fact,power;

printf("Enter value of x (in radians): ");
```

```
scanf("%f",&x);
printf("Enter number of terms: ");
scanf("%d",&terms);
for(i=1;i<=terms;i++)
{
fact=1;
power=pow(x,(2*i-1));
for(int j=1;j<=(2*i-1);j++)
fact=fact*j;
sum=sum+sign*(power/fact);
sign=-sign;
}
printf("Sum of sine series = %.6f",sum);
return 0;
}
```

INPUT: Value of x and number of terms.

```
Enter value of x (in radians): 70
Enter number of terms: 2
```

OUTPUT: Displays the sum of the sine series.

```
Sum of sine series = -57096.667969
```

```c
#include <stdio.h>
#include <math.h>
int main()
{
    int terms, i, sign = 1;
    float x, sum = 0, fact, power;
    printf("Enter value of x (in radians): ");
    scanf("%f", &x);
    printf("Enter number of terms: ");
    scanf("%d", &terms);
    for(i = 1; i <= terms; i++)
    {
        fact = 1;
        power = pow(x, (2 * i - 1));
        for(int j = 1; j <= (2 * i - 1); j++)
        {
            fact = fact * j;
        }
        sum = sum + sign * (power / fact);
        sign = -sign;
    }
    printf("Sum of sine series = %.6f", sum);
    return 0;
}
```

SAMPLE INPUT:
Enter value of x (in radians): 1
Enter number of terms: 5

SAMPLE OUTPUT:
Sum of sine series = 0.841471

CONCLUSION: Thus, the C program successfully calculates the sum of the sine series for the given number of terms.

FAQs:

1. What is a sine series in mathematics?
2. Why is factorial used in the sine series?
3. Why does the sine series have alternating signs?

4. Which header file is required for power functions in C?
5. How does increasing the number of terms affect accuracy?